

Production Readiness Report

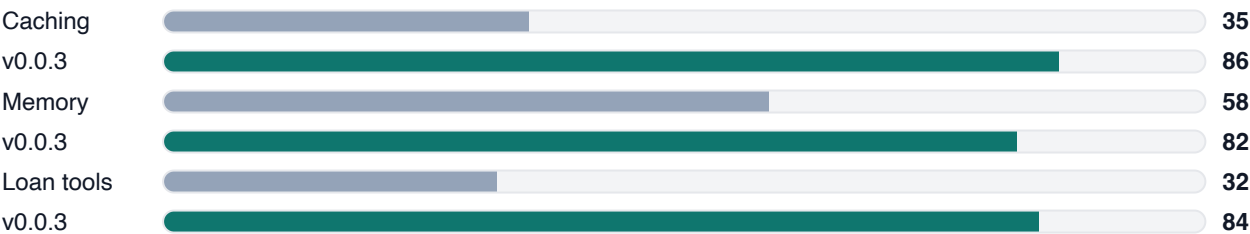
ANAN Agent - New Version Technical Report

Date: February 17, 2026 | Scope: Backend agent capability upgrade | Commit: 364ec45

0) Version Comparison (0.0.2 vs 0.0.3)

Capability	v0.0.2	v0.0.3	Delta
Cross-user cache reuse	No shared cache	Global shared cache with hit tracking	High improvement
Search response consistency	Per-user only	Global + per-user layered retrieval	Improved consistency
Loan support	General bank info	Loan search + installment calculator	New capability
Long-conversation memory	Basic extraction	Timeline/financing/contact/name variants	Better recall

Relative Capability Score (0-100)



1) Core Upgrades Delivered

- Global cross-user property search cache with TTL and hit tracking.
- Smart search pipeline now reuses shared cache before per-user cache.
- Refresh behavior preserved: refresh/bypass does not reuse stale cache.
- New loan capability tools: live loan search + mortgage calculator.
- Improved memory extraction for long conversations and user recall.

2) Performance/Scale Impact

- Reduces duplicate web search calls across users with same query intent.
- Improves latency on repeated intents via cache-first retrieval.
- Lowers token and search cost by reusing normalized result payloads.
- Supports high-concurrency query stacks with deterministic cache keys.

3) Architecture Changes

Area	Change	Outcome
Schema	<code>globalSearchCache</code> table added.	Shared cross-user cache persistence.
Service APIs	<code>getGlobalSearchCache</code> , <code>upsertGlobalSearchCache</code> , <code>trackGlobalSearchCacheHit</code> , <code>clearGlobalSearchCache</code> .	Cache lifecycle control + admin invalidation.
Property Tool Flow	<code>smartPropertySearch</code> now checks global cache, then user cache, then web fallback.	Faster repeated searches, fewer external calls.
Loan Tooling	<code>searchSaudiLoans</code> + <code>calculateSaudiLoan</code> added.	End-to-end financing assistance in-agent.
Memory Inference	Added extraction for timeline, financing mode, contact preference, and name variants.	Better long-session personalization and recall.

4) Validation Status

- Tests passed for memory and tool flows (Vitest targeted suites).
- TypeScript typecheck passed: `tsc --noEmit` .
- No new regression failures detected in modified modules.

5) Known Constraints

- Repository-wide lint has existing unrelated legacy issues.
- Git push blocked by remote permission (HTTP 403 write denied).
- Commit exists locally and is ready for authorized push.

6) Production-Ready Recommendation

- Keep this version as the new baseline for search/memory orchestration.
- Add dashboard visibility for global cache hit ratio and freshness misses.
- Add explicit user command for forced refresh/invalidation per query.
- Run full lint remediation in a separate cleanup track (not in feature branch).